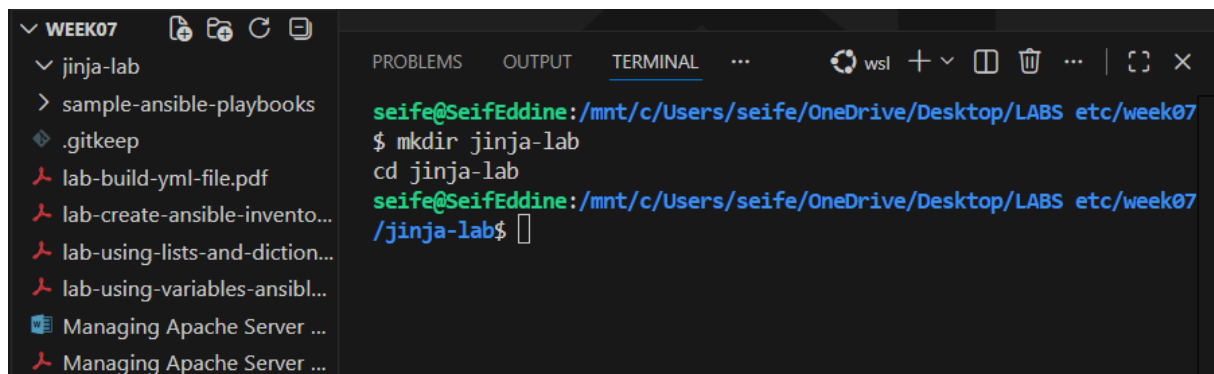
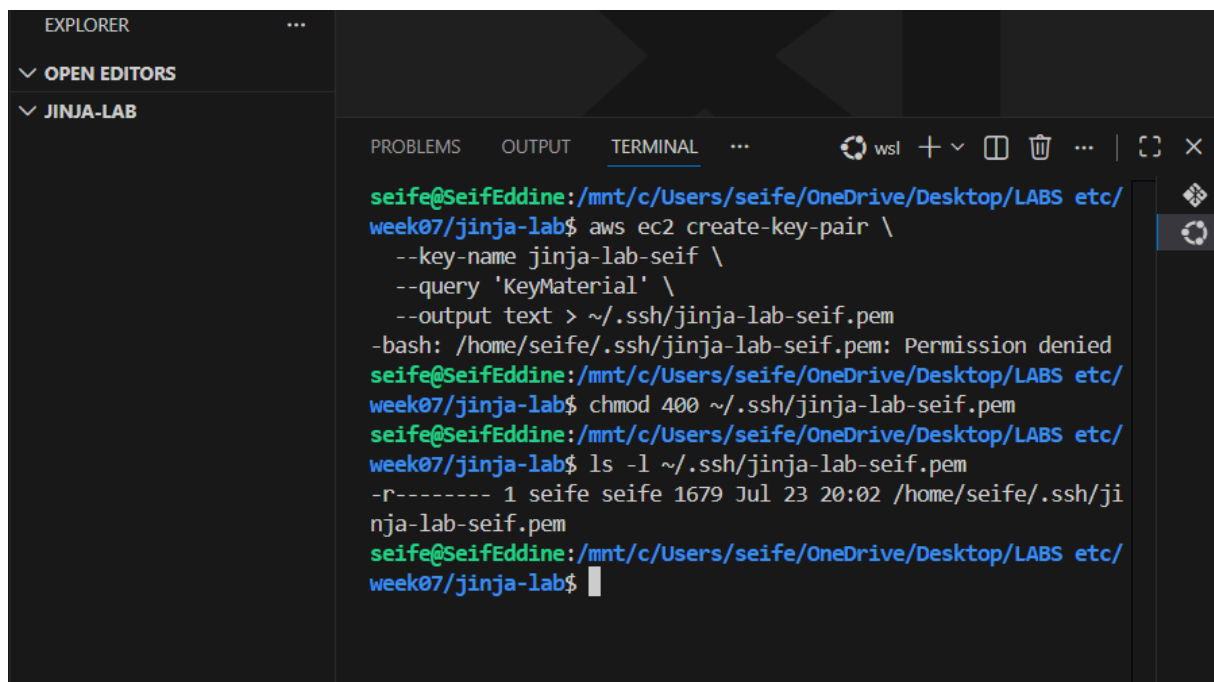


lab-setup-remote-servers



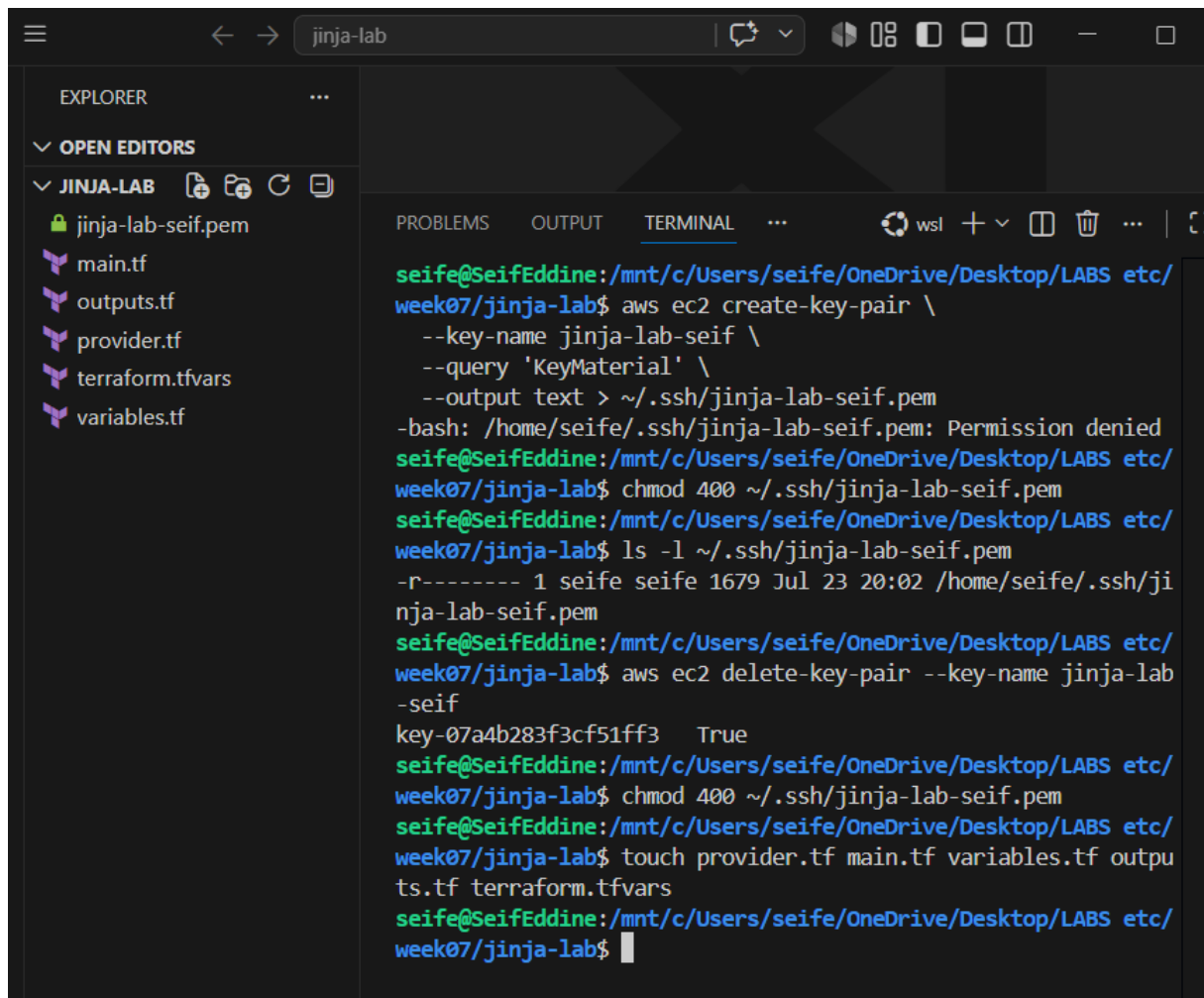
The screenshot shows a VS Code interface with a sidebar on the left containing a file explorer. The main area is a terminal window. The terminal shows the user creating a directory named 'jinja-lab' and navigating into it.

```
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07
$ mkdir jinja-lab
cd jinja-lab
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07
/jinja-lab$
```



The screenshot shows a VS Code interface with a sidebar on the left containing a file explorer. The main area is a terminal window. The terminal shows the user running the 'aws ec2 create-key-pair' command to create a key pair named 'jinja-lab-seif'. It then shows an error message 'Permission denied' when trying to access the key file, followed by the user running 'chmod 400' to set permissions. Finally, the user runs 'ls -l' to verify the permissions.

```
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ aws ec2 create-key-pair \
--key-name jinja-lab-seif \
--query 'KeyMaterial' \
--output text > ~/.ssh/jinja-lab-seif.pem
-bash: /home/seife/.ssh/jinja-lab-seif.pem: Permission denied
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ chmod 400 ~/.ssh/jinja-lab-seif.pem
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ ls -l ~/.ssh/jinja-lab-seif.pem
-r----- 1 seife seife 1679 Jul 23 20:02 /home/seife/.ssh/ji
nja-lab-seif.pem
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$
```



```
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ aws ec2 create-key-pair \
--key-name jinja-lab-seif \
--query 'KeyMaterial' \
--output text > ~/.ssh/jinja-lab-seif.pem
-bash: /home/seife/.ssh/jinja-lab-seif.pem: Permission denied
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ chmod 400 ~/.ssh/jinja-lab-seif.pem
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ ls -l ~/.ssh/jinja-lab-seif.pem
-r----- 1 seife seife 1679 Jul 23 20:02 /home/seife/.ssh/ji
nja-lab-seif.pem
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ aws ec2 delete-key-pair --key-name jinja-lab
-seif
key-07a4b283f3cf51ff3    True
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ chmod 400 ~/.ssh/jinja-lab-seif.pem
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$ touch provider.tf main.tf variables.tf outpu
ts.tf terraform.tfvars
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/
week07/jinja-lab$
```

```
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/
jinja-lab$ terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.56.0...
- Installed hashicorp/aws v6.56.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the p
rovider
selections it made above. Include this file in your version control r
epository
so that Terraform can guarantee to make the same selections by default
when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan
" to see
any changes that are required for your infrastructure. All Terraform
commands
should now work.

If you ever set or change modules or backend configuration for Terraf
orm,
rerun this command to reinitialize your working directory. If you for
get, other
commands will detect it and remind you to do so if necessary.
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/
jinja-lab$ █
```

```

seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/
jinja-lab$ terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.56.0...
- Installed hashicorp/aws v6.56.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the p
rovider
selections it made above. Include this file in your version control r
epository
so that Terraform can guarantee to make the same selections by defaul
t when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan
" to see
any changes that are required for your infrastructure. All Terraform
commands
should now work.

If you ever set or change modules or backend configuration for Terraf
orm,
rerun this command to reinitialize your working directory. If you for
get, other
commands will detect it and remind you to do so if necessary.
seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/
jinja-lab$ terraform validate
Success! The configuration is valid.

seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/
jinja-lab$ █

```

Ln 7, Col 2 Spaces: 4 UTF-8 CRLF { } Plain Text Sign In 🔔

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ public_dns = [
  + (known after apply),
  + (known after apply),
]
+ public_ips = [
  + (known after apply),
  + (known after apply),
]

```

```
aws_instance.web[0]: Creating...
aws_instance.web[1]: Creating...
aws_instance.web[1]: Still creating... [00m13s elapsed]
aws_instance.web[0]: Still creating... [00m13s elapsed]
aws_instance.web[0]: Creation complete after 16s [id=i-09c50410539248579]
aws_instance.web[1]: Creation complete after 16s [id=i-0d295385608b50558]
```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

```
public_dns = [
  "ec2-63-178-10-228.eu-central-1.compute.amazonaws.com",
  "ec2-3-65-198-148.eu-central-1.compute.amazonaws.com",
]
public_ips = [
  "63.178.10.228",
  "3.65.198.148",
]
```

ja-lab\$

The authenticity of host '18.159.41.162 (18.159.41.162)' can't be established.

ED25519 key fingerprint is SHA256:wsprmI80ShFkcXWmK/HoSQKglSMI8AJxe0Yzyx7vdzM.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? 3.6

8.198.250 | SUCCESS => {

"ansible_facts": {

"discovered_interpreter_python": "/usr/bin/python3"

},

"changed": false,

"ping": "pong"

}

18.159.41.162 | SUCCESS => {

"ansible_facts": {

"discovered_interpreter_python": "/usr/bin/python3"

},

"changed": false,

"ping": "pong"

}

seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/jin

ja-lab\$

3.68.198.250 | CHANGED => {

"ansible_facts": {

"discovered_interpreter_python": "/usr/bin/python3"

},

"cache_update_time": 1784835363,

"cache_updated": true,

"changed": true,

"stderr": "",

"stderr_lines": [],

"stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following additional packages will be installed:\n nginx-common\nSuggested packages:\n fcgiwrap nginx-doc ssl-cert\nThe following NEW packages will be installed:\n nginx nginx-common\n0 upgraded, 2 newly installed, 0 to remove and 28 not upgraded.\nNeed to get 568 kB of archives.\nAfter this operation, 1602 kB of additional disk space will be used.\nGet:1 http://eu-central-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.15

```
PROBLEMS OUTPUT TERMINAL ... wsl ! + v [ ] [ ] ... [ ] [ ] x

seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/jinja-lab$

PLAY [web] *****
*****

TASK [Gathering Facts] *****
*****
ok: [18.159.41.162]
ok: [3.68.198.250]

TASK [Copy Jinja2 template] *****
*****
changed: [18.159.41.162]
changed: [3.68.198.250]

PLAY RECAP *****
*****
18.159.41.162      : ok=2    changed=1    unreachable=0    failed=0
d=0    skipped=0    rescued=0    ignored=0
3.68.198.250      : ok=2    changed=1    unreachable=0    failed=0
d=0    skipped=0    rescued=0    ignored=0

seife@SeifEddine:/mnt/c/Users/seife/OneDrive/Desktop/LABS etc/week07/jinja-lab$
```

